SOLUTION REPORT

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Industry: Healthcare
Project Title: Improving Maternal Health Insights Through Data Analysis
Proposed Approach: Python-Based Analysis of Maternal Health Data

1. Project Overview

The Maternal Health Analysis Project examines maternal healthcare utilisation in Nigeria, focusing particularly on place of delivery and its relationship with women's demographic and socioeconomic characteristics.

The primary objective is to understand where women deliver their babies and how delivery patterns differ according to factors such as age, place of residence, educational attainment, and household wealth. The project also examines antenatal care utilisation and Caesarean section deliveries.

The analysis aims to identify patterns and potential inequalities in maternal healthcare utilisation that can provide useful insights for further research and maternal health interventions.

2. Data Source

The project uses secondary data from the **2024 Nigeria Demographic and Health Survey (NDHS)**. The dataset was obtained through the NDHS portal and provided in Stata (.DTA) format.

The specific file used was NGIR8BFL.dta, an individual recode dataset. The original dataset contained 39,050 records and 6,411 variables. Due to the large number of variables, relevant demographic and maternal healthcare indicators were identified and extracted for the analysis.

The selected variables included age, residence, education, household wealth, tetanus injections, antenatal visits, place of delivery and Caesarean section status.

3. Methodology and Steps Followed

The project followed a structured data analysis workflow using Python and relevant statistical libraries.

a. **Data loading and inspection:** The Stata dataset was imported using Pandas. Its dimensions, variable names, data types and general structure were examined to understand the dataset and identify variables relevant to the research objectives.

b. **Data cleaning and preparation:** Relevant variables were extracted into a smaller working dataset without modifying the original dataset. Missing-value patterns were investigated before removing or modifying records. Numerical codes were interpreted using the dataset's variable labels and value labels. Special codes representing "Don't know" responses were converted to missing values, while coded variables were recategorized into readable labels. Redundant raw columns were removed after their cleaned equivalents had been created. The resulting maternal analysis dataset contained 13,968 records. The core variables for antenatal visits, delivery location and Caesarean section had available responses in these records, although some special "Don't know" codes were subsequently treated as missing.

c. **Exploratory data analysis:** Descriptive statistics and frequency distributions were generated to examine respondents' age, residence, educational attainment, household wealth and antenatal visits. Preliminary comparisons of delivery location across demographic groups were also conducted.

d. **DHS survey design specification:** The complex survey structure was incorporated using the women's sampling weight (v005), primary sampling unit (v021) and sampling-error stratum (v022). The weights were divided by one million as required for DHS weight scaling. The survey design variables were validated, with 1,368 unique clusters and 74 unique strata identified.

e. **Survey-weighted statistical analysis:** The Samplics Python library (version 0.6.0) was used to calculate weighted population proportions and means using Taylor-series variance estimation. The analysis accounted for the survey design and used eligibility domains on the full survey dataset to preserve the original survey structure. Estimates included standard errors and 95% confidence intervals.

f. **Data visualisation:** Matplotlib was used to produce five visualisations showing overall delivery location, delivery by residence, delivery by education, delivery by household wealth and the mean number of antenatal visits with its 95% confidence interval.

4. Key Observations and Findings

The survey-weighted analysis revealed the following patterns:

- **Place of delivery:** Home delivery was the most common, accounting for 57.2% of deliveries. Public facilities accounted for 27.0%, private facilities 14.4%, NGO facilities 0.9% and other locations 0.5%.
- **Residence:** Home delivery was more prevalent among rural women (71.4%) than urban women (34.3%). Urban women recorded higher proportions of public facility deliveries (37.8%) and private facility deliveries (25.4%).
- **Educational attainment:** Home delivery declined as educational attainment increased, from 82.2% among women with no education to 9.6% among women with higher education. Public and private facility deliveries became more prevalent with increasing educational attainment.
- **Household wealth:** Home delivery declined from 83.0% among women in the poorest wealth group to 11.0% among those in the richest group. Private facility delivery increased considerably, from approximately 1.9% to 42.6%.
- **Antenatal care:** The survey-weighted mean number of antenatal visits was 4.31, with a 95% confidence interval of 4.16–4.45 visits.
- **Caesarean section:** Approximately 5.3% of deliveries were by Caesarean section, while 94.7% were not.

5. Analysis and Conclusion

The findings reveal substantial differences in maternal healthcare utilisation across residential and socioeconomic groups. Home delivery was considerably more common among rural women, women with lower educational attainment and women from poorer households. Conversely, facility-based deliveries were more prevalent among urban, more educated, and wealthier women.

These patterns highlight the importance of considering socioeconomic circumstances and residential differences when examining maternal healthcare utilisation. Financial constraints, geographical accessibility and educational opportunities are potential areas for further investigation, although the descriptive findings alone cannot establish these factors as causes of the observed differences.

The incorporation of DHS sampling weights, clusters and strata strengthened the analysis by accounting for the complex survey design. Overall, the project demonstrates how secondary health data can be cleaned, analysed and visualised to identify important patterns in maternal healthcare utilisation.

6. Recommendations and Further Work

The project can be extended through the following activities:

1. **Survey-adjusted regression analysis:** Apply appropriate survey-weighted logistic or multinomial regression models to investigate factors associated with home and facility delivery.
2. **Geographical analysis:** Examine differences in delivery patterns across regions or states to identify areas with potentially lower utilisation of facility-based maternal healthcare.
3. **Further antenatal care analysis:** Investigate how antenatal visits vary across residence, education, and household wealth groups.
4. **Survey-adjusted statistical testing:** Conduct appropriate hypothesis tests to determine whether observed differences between demographic groups are statistically significant while accounting for the DHS survey design.
5. **Missing-data investigation:** Further examine missing observations and special response codes, particularly for antenatal visits and tetanus-related information.

6. **Policy-oriented recommendations:** Use the findings to inform further investigation into strategies for improving access to affordable, accessible, and quality maternal healthcare, particularly among rural and economically disadvantaged women.

Conclusion: The Maternal Health Analysis Project successfully applied data cleaning, exploratory analysis, complex survey estimation and data visualisation to the 2024 Nigeria DHS dataset. The findings highlight notable differences in place of delivery across residence, education, and household wealth groups. The project provides a useful foundation for further statistical investigation and evidence-informed approaches to improving maternal healthcare utilisation.